

SRIRAM DEVATA

Champaign, IL 61820, USA

☎ +1 447 902 7132 ✉ sdevata2@illinois.edu, sriramdvt@gmail.com 🌐 sriramdvt.github.io

Education

University of Illinois at Urbana-Champaign (UIUC)

Aug. 2024 – May 2029

PhD in Computer Science, advised by Prof. Sarita Adve

3.83/4

International Institute of Information Technology, Hyderabad (IIITH)

Bachelor of Technology (Honours) in Computer Science and MS by Research

CGPA - 8.99/10

Research/Work Experience

University of Illinois at Urbana-Champaign (UIUC)

Aug. 2024 - Present

Graduate Research Assistant

- My broad research focus is to build software and hardware for datacenters.
- Developed a serving system for managing datacenter GPUs that decreases the amount of GPU resources needed to serve compound AI inference workloads by more than 2x.

STARLab, Carnegie Mellon University (CMU)

Jan. 2022 - June 2024

Student Researcher

- Worked with Prof. Akshitha Sriraman and Dr. Gilles Pokam to identify and leverage hardware design opportunities to enhance efficiency in datacenters by optimizing redundant datacenter “tax” operations.
- Developed a methodology that uses Linux perf to profile widely-used open-source microservices to uncover patterns that can guide the development of general hardware accelerators.

Chaincode Labs, Summer of Bitcoin

July 2021 - Jan 2022

Software Developer (Independent Contractor)

- Streamlined the testing operations of Bitcoin Core by replacing the compiled wallet with a versatile Python mini-wallet.
- I spearheaded the development of a ‘-nomempool’ runtime option. This feature allows for the operation of a Bitcoin node without a mempool, allowing users to participate in the Bitcoin network without needing to mine blocks.

Relevant Publications

JigsawServe: Compound Inference Serving on Datacenter GPUs

October 2025

Under Review

Sriram Devata, Rahul Singh, Sarita Adve

JigsawServe is the first inference serving system to jointly optimize for latency, accuracy, and fine-grained cost for serving a compound inference system. Across many evaluation scenarios, JigsawServe consumes only 43.3% of the available GPU resources on average while meeting accuracy SLOs with less than 0.6% latency SLO violations.

Technical Skills

Languages/Frameworks: Python, C++, CUDA, PyTorch

Utilities/Technologies: Linux, Git, Bash, Slurm, Jupyter, Docker

Grants, Resources, and Funding

Serving Compound AI Workloads on Datacenter GPUs

2025

Received resources with an estimated value of \$245,510 through the NSF ACCESS program as a PI.

Awards and Honours

Qualcomm Innovation Fellowship Finalist

2025

W.J. Poppelbaum Memorial Award (for graduate students in computer architecture), *UIUC*

2025

Student Travel Grant, *MLSys'25*

2025

CS PhD Fellowship, *UIUC*

2024

MITACS Globalink Research Internship, *University of Waterloo*

2023

Dean's Merit List for Academic Excellence, *IIIT Hyderabad*

2019-2021

Relevant Coursework

- Manycore Parallel Algorithms (UIUC)
- Compiler Construction (UIUC)
- Distributed Systems (UIUC)
- Advanced Topics in Computer Architecture (UIUC)
- Parallel Computer Architectures (UIUC)
- Introduction to Software Systems (IIIT-H)
- Topics in Deep Learning (IIIT-H)
- Advanced NLP (IIIT-H)
- Operating Systems and Networks (IIIT-H)
- Design and Analysis of Software Systems (IIIT-H)

Selected Course Projects

GPU Spatial Partitioning for Multi-model Workloads

Jan - May 2025

We evaluated how spatial partitioning on GPUs would impact the working of ML inference pipelines. We found that there is very little overhead due to the additional data movement between spatial partitions, and there is a great advantage in the ML inference latencies being consistent and predictable.

XR Workload Scheduler on a Mobile Device

Jan - May 2025

We developed multiple schedulers that choose to run ML inference on the CPU/GPU/NPU of a mobile XR device to minimize the overall queuing and inference latency. In terms of number of deadlines missed, two of our schedulers perform at least 6x better than using only one of the CPU/GPU/NPU.

Distributed File System

Aug - Dec 2024

Implemented a replicated, distributed file system in Golang. The filesystem is resilient and adaptable by implementing features such as heartbeat messages to maintain membership and replication, caching to minimize network traffic, and chunking of large files into smaller blocks.

Desh: a Custom Linux Shell in C

Oct 2021

‘Desh’ is a custom shell designed for Linux-based systems developed from scratch in C, relying exclusively on system calls. This approach required a comprehensive understanding of Linux system architecture and proficient command over C programming, showcasing advanced skills in both system-level programming and operating system concepts.

Professional/Academic Service

Operating Systems Design and Implementation (OSDI'25) Artifact Evaluation Committee

2025

Machine Learning and Systems (MLSys'25) Artifact Evaluation Committee

2025

Compilers, Architecture, and Parallel Computing (CAP) Seminar Organizing Committee, *UIUC*

2025-Present

Other Publications

Generative artificial intelligence for small molecule drug design

August 2024

Current Opinion in Biotechnology

Ganesh Chandan Kanakala*, Sriram Devata*, Prathit Chatterjee, Deva Priyakumar

TorRNA - Improved Prediction of Backbone Torsion Angles of RNA by Leveraging Large Language Models

June 2024

ChemRxiv

Sriram Devata, Deva Priyakumar

DeepSPInN - Deep reinforcement learning for molecular Structure Prediction from Infrared and ¹³C NMR spectra

March 2024

Digital Discovery, Royal Society of Chemistry

Sriram Devata*, Bhuvanesh S.*, Sarvesh M.*, Yashaswi P. , Siddhartha L., Girish V. , Deva Priyakumar